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Competitor actions, customer integration, and supply chain responsiveness: A contingency–capability-based view

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ABSTRACT

Conventional wisdom suggests that market forces such as competitor aggressiveness and competitor innovativeness induce supply chain responsiveness. However, this assertion does not only lack empirical evidence but also the mechanisms that explain the supply chain responsiveness effects of market forces are under-theorized. Drawing on the contingency–capability perspective, this research develops and tests the argument that customer integration is a critical boundary-spanning capability that translates competitor aggressiveness and innovativeness into enhanced supply chain responsiveness. Empirical results based on survey data from 117 firms in Ghana reveal that both market forces are not directly related to supply chain responsiveness. Additional results, however, show that customer integration mediates the competitor aggressiveness and innovativeness–supply chain responsiveness relationships. In contributing to the limited knowledge of the determinants of responsive supply chains, this article shows that external environmental factors are essential but might be insufficient for accounting for the heterogeneity in supply chain responsiveness.

KEYWORDS

Customer integration; supply chain responsiveness; competitor aggressiveness and innovativeness; contingency–capability-based view; Ghana

Introduction

The global market of the 21st Century is characterized by high saturation, business environment uncertainties, and turbulences that have further intensified the competitive dynamics for modern businesses (Yu et al., 2019, 2020). Under such market conditions, scholars argue that supply chain responsiveness is not only desirable but also critical for driving competitive advantage and business survival (Essuman et al., 2021; Yang et al., 2019). Accordingly, there is a growing assumption that supply chain responsiveness emerges within market environments that trigger turbulence and uncertainties (Essuman et al., 2021; Gunasekaran et al., 2008; Nenavani & Jain, 2022; Yu et al., 2020). Despite the importance of supply chain responsiveness, it is rarely studied in the context of competitor actions (Ibrahim & Harrison, 2020). Supply chain responsiveness refers to the degree to which a firm's supply chain fulfills diverse customer requirements in a time-effective manner (Essuman et al., 2021; Gunasekaran et al., 2008).

There is a recognition that competitor actions in terms of the degree of aggressiveness and

innovativeness constitute a set of intentional and purposeful initiatives that consistently and significantly disturb market conditions in terms of customer preferences and requirements (Jajja et al., 2018; Mckinley et al., 2014). Recent studies, however, suggest that competitor actions tend to have a “threat – opportunity” paradox effect on rival firms (Hutzschenreuter et al., 2021; Moreira et al., 2020). The few studies examining competitor actions have shared insights on how focal firms offset such uncertainties in defense of their market position. Turner et al. (2010) show that competitor innovativeness is responded to by increasing generational product innovation in the software product lines in the United States of America. Peng et al. (2017) show that competitors' activities spur insurers' bancassurance strategies in Taiwan. Also, it is shown that pharmaceutical firms respond to competitors' new product launches by increasing technological licensing-in (Moreira et al., 2020). On the performance impact of competitor actions, Ibrahim and Harrison (2020) found competitor actions to hurt profitability among UK car manufacturers. More recently, Hutzschenreuter et al. (2021) advance that competitor actions such as product promotion and

product quality intensification tend to increase the opportunity for growth in the product market. Apparently, the threat-opportunity paradox in the environmental uncertainty literature remains unresolved (Ibrahim & Harrison, 2020; Yu et al., 2020).

The literature indicates that focal firms' reaction to competitor actions depends on their correct interpretation of competitive actions as either a threat or an opportunity (Hutzschenreuter et al., 2021), which in turn calls for real-time visibility of competitor initiatives (Setiawan et al., 2015). Furthermore, in a time-based intense competition, a swift response to competitive actions is critical (Reichhart & Holweg, 2007). Impliedly, while successfully responding to the rapid-paced competitor actions is key for maintaining a competitive position, it can overburden the internal resources, capabilities, and competencies of the focal firm (Moreira et al., 2020). This suggests that to respond effectively to competitor actions, firms need to engage in integration initiatives with supply chain partners (Fianko et al., 2022; Moyano-Fuentes et al., 2016). Customer integration, given its customer-market centeredness, is particularly crucial in this regard. Specifically, customer integration allows firms to appreciate customers' changing requirements in real-time, reduce structural and relational gaps between the firm and target customers, and permit the firm to co-produce and deliver products and services rapidly (Fianko et al., 2022). To cope with competition while increasing supply chain responsiveness, successful companies such as Dell, Amazon, and Walmart have over the years invested heavily in customer integration initiatives (Min et al., 2019). Therefore, this research seeks to develop and test the contention that customer integration is an important boundary-spanning capability that enables firms to achieve superior supply chain responsiveness in the face of growing competitive actions.

On this front, this research answers two questions: (1) *how do competitor actions (including aggressiveness and innovativeness) relate to supply chain responsiveness?* (2) *does customer integration mediate the relationships between competitor actions and supply chain responsiveness?* By analyzing these questions, we contribute to literature and practice as follows. First, we contribute to the conceptual literature on the interface between supply chain responsiveness and external environment factors (Ibrahim & Harrison, 2020; Nenavani & Jain, 2022). We specifically interrogate and offer an improved understanding of the assumed links between competitor actions and supply chain responsiveness. Contrary to normative assumptions

(e.g., Essuman et al., 2021; Nenavani & Jain, 2022), results from this study indicate that supply chain responsiveness does not necessarily result from competitor actions. Rather, and consistent with our theorization, we find that, as competitor actions increase, it requires firms with strong customer integration to attain superior supply chain responsiveness (Fianko et al., 2022; Moyano-Fuentes et al., 2016). Second, we expand the limited literature on the external environment-related antecedents of customer integration (Yu et al., 2020). We identify competitor aggressiveness and competitor innovativeness as significant drivers of customer integration. Third, we demonstrate how external environment variables (i.e., competitor aggressiveness and competitor innovativeness) and one internal environment variable (i.e., customer integration) combine to determine levels of supply chain responsiveness. The desirability of and the normative assumption that supply chain responsiveness is an automatic outcome of market conditions that trigger turbulence (Nenavani & Jain, 2022; Yu et al., 2020) while restraining scholarly efforts to understand the critical drivers of supply chain responsiveness, can undermine firms' efforts toward and successes in achieving supply chain responsiveness. That said, this study details how in the face of increasing competitor actions, firms can deploy customer integration to enhance supply chain responsiveness. The rest of the article is composed of the theoretical background and hypotheses, methods & results, discussions & implications, limitations, and avenues for further research.

Theoretical background and hypotheses

To investigate our research questions, we draw on and blend insights from the contingency theory and the capability-based view. Notably, organizations, and for that matter, supply chains, operate in open systems, which are complex, unstable, and unpredictable (Blackhurst et al., 2011). As such, it is insufficient to rely on any single strategy in any given environment to achieve sustainable outcomes (Johnson et al., 2001; Williams et al., 2017). Accordingly, the contingency theory suggests that organizations would normally align their internal processes, structures, strategies, etc. with the prevailing external environment factors to obtain 'fit', and enhance performance (Donaldson, 2006; Fawcett & Scully, 1995; Johnson et al., 2001). Largely, the external environment is viewed as 'firm-independent'; consisting of factors that can influence organizations' success (Islam & Hu, 2012). Therefore, for firms to excel, they need to, from time to time,

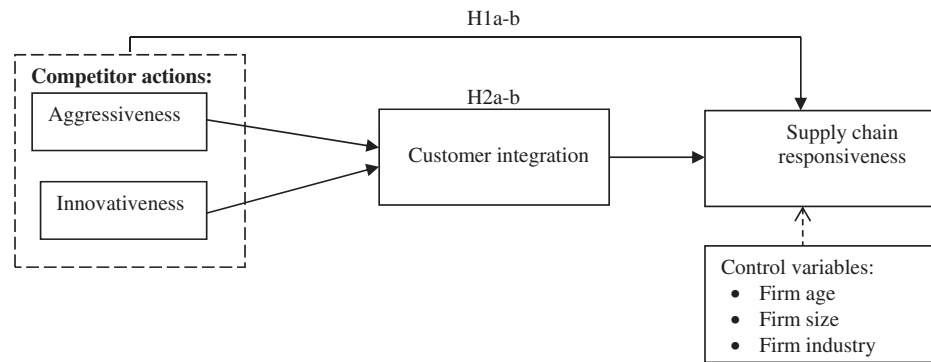


Figure 1. Research model.

restructure their operations and strategies to cope and adapt to the changes in the external environment to achieve fit (Donaldson, 2006; Grötsch et al., 2013).

From the contingency theory perspective, Johnson et al. (2001) and Simangunsong et al. (2012) assert that environment turbulence factors (i.e., sources of uncertainties) affect supply chain strategic options (i.e., uncertainty coping and reduction strategies) which in turn affect supply chain performance. Also, Fawcett and Scully (1995) illustrate the effects of competitive intensity and the global market on supply chain performance outcomes via just-in-time practices. Williams et al. (2017) argue that both competition and adaptability are central to the contingency theory which explain environment-structure/strategy-performance linkages. Thus, adaptability reflects the capabilities that firms require to deal with unforeseen events, turbulence, disruptions, and shocks that might result from competitor aggressiveness and innovativeness (Stolker et al., 2008). As Eckstein et al. (2015) intimate, it is necessary for firms operating in turbulent environments to develop capabilities to adapt their supply chains to cope and become responsive in addressing market needs.

Given this, we consider the infusion of the capability-based view into the contingency theory as relevant to addressing our second research question. The capability-based view considers organizations as possessing a blend of resources and competencies that could provide a source of competitive advantage and enhanced performance; provided the resources and competencies available are valuable, rare, inimitable, and non-substitutable (Blackhurst et al., 2011; Brandon-Jones et al., 2014). Resources comprise both physical assets (e.g., IT and communication infrastructure, materials) and intangible assets (i.e., goodwill, knowledge) whereas competencies are the skills and abilities by which resources are utilized effectively via an organization's processes and activities (Barney, 1991; Johnson et al., 2008). Competencies are

'operants' necessary to optimize resource benefits (Adams et al., 2014). Capabilities are possessed by and reside within firms. Through integration, supply chain partners can pull together and share active and dormant capabilities, resources and competencies scattered within the network to enhance supply chain performance (Annan et al., 2016; Leuschner et al., 2013).

As prior research points out, effectively adjusting to, and coping with environment turbulence require developing response mechanisms in the forms of capabilities and strategies (Flynn et al., 2016). From a supply chain perspective, Fawcett and Scully (1995) indicate that, in an intensified competitive environment, survival requires that managers adopt supply chain strategies that help their firms enhance supply chain outcomes. As shown in Figure 1, we argue that competitor actions manifesting in the forms of aggressiveness and innovativeness could be a driving force for firms to develop boundary-spanning capabilities such as customer integration (Adams et al., 2014; Moyano-Fuentes et al., 2016) to enhance their supply chain responsiveness (Fianko et al., 2022; Flynn et al., 2016).

Competitor actions and supply chain responsiveness

Competitor actions encapsulate strategic, tactical, and or operational moves orchestrated by firms in a given market space to undo their competitors, win orders, and capture the market (Ibrahim & Harrison, 2020). Such moves include but are not limited to innovation activities, promotional activities (e.g., advertisement, pricing, discount) and product offerings (e.g., variety, quality) (Ibrahim & Harrison, 2020). This study focuses on two generic competitive actions: innovativeness and aggressiveness. Competitor aggressiveness reflects the degree to which competitor actions are geared toward outwitting other competing firms, usefully in a hostile manner (Jajja et al., 2018).

Thus, competitor aggressiveness manifests in unhealthy strategic marketing moves (e.g., competitors focusing on crude reverse-engineering practices) and unfriendly marketing campaigns (e.g., fierce price cuts, frequent sales promotion leading to price wars). On the other hand, competitor innovativeness depicts the rate at which competitors introduce new products and processes into the market (Mckinley et al., 2014). Fundamentally, increasing levels of competitor aggressiveness and innovativeness result in market turbulence and uncertainty (Jajja et al., 2018).

We argue that competitor actions in the business environment can lead to improved supply chain responsiveness. Competitors' behavior can induce turbulence and uncertainty in the competitive environment which have a profound effect on the focal firm's activities, especially its responsiveness to market needs. In particular, changes in the competitive environment determine the level of the focal firm's need for responsiveness; such that an increase in the levels of competition and market changes require firms to improve their responsiveness in order to survive (Luu, 2017). As competitor aggressiveness and innovativeness continue to induce turbulence and volatility within the business environment, focal firms are more likely to accelerate their activities to respond quickly and effectively to market changes (Luu, 2017). Specifically, and in line with the mantra that "desperate circumstances can call for desperate measures" (Covin & Covin, 1990), firms that are confronted with aggressive marketing campaigns and continuous introduction of innovative products from competitors into the market are expected to counteract such actions. In particular, firms are more likely to adjust their resources and streamline logistics activities to secure a first-mover advantage through actions such as a flexible operational approach, improved delivery speed, order fulfillment, and reduced lead time. Thus, competitor actions compel firms to be decisive players in a competitive environment by acting forcefully to improve responsiveness to market needs (Stambaugh et al., 2020) and maintain a competitive position. Accordingly, we argue that increases in competitor actions are expected to be matched with increases in supply chain responsiveness.

Competitors' aggressive operational activities are intended to dilute the focal firm's competitive advantage in the market, woo customers (actual and potential) of the focal firm and drive profitability downward (Hutzschenreuter et al., 2021). Under such scenarios, what successful firms seek to do is to stay ahead of competitors by upping their game plan.

Sensing the threats from rivals, the focal firm may respond by varying and adapting their product/service, and competitive strategies, to distinguish their offerings further from their peers (Hutzschenreuter et al., 2021). Also, the focal firm may counteract rivals' aggressive actions by increasing their product-to-market speed and responding quickly to customers' implied and stated demands and increasing their aggressive marketing strategies (Peng et al., 2017). That is, in the focal firm's attempt to react to competitors' aggressive behaviors, they develop the capacity to improve their responsiveness to the emerging and changing trends in the market in the processes of rapid continuous learning, research and development (R&D), innovative activities, market penetration, etc. Consequently, we hypothesize that

H1a: Competitor aggressiveness is positively associated with supply chain responsiveness.

Where competitors engage in rapid process improvement, frequent new product development and launch, and service improvements, they intend to enhance their competitive edge by way of rendering the focal firm's processes and products/services obsolete (Moreira et al., 2020). Hutzschenreuter et al. (2021), however, contend that when competitors engage in innovativeness, they signal trends of future development in the product/service market. As competitors reveal their intentions, the focal firm can take advantage of this knowledge to equally engage in more directed research and development initiatives; that is, intensifying their own innovativeness to enable them to stay ahead of the competition (Moreira et al., 2020). To illustrate, incumbent players in the automotive industry (such as General Motors) had demonstrated the potential for commercializing electric vehicles (EVs) earlier on but were not motivated to carry on this intention. Later, a new entrant, Tesla, has surpassed the incumbents in the market of EVs, in the era of the climate-carbon emergency (Thomas & Maine, 2019). Thus, the focal firm can increase its supply chain responsiveness in the market by learning from the innovation signals of rivals. Accordingly, we propose that:

H1b: Competitor innovativeness is positively associated with supply chain responsiveness.

The mediation role of customer integration

We propose customer integration as one of the critical generative processes that link competitive actions to supply chain responsiveness. Customer integration reflects the degree to which a firm strategically

collaborates with its customers (Flynn et al., 2010) through information exchanges, collaborative decision-making, and process synchronization (Munir et al., 2020; Shou et al., 2018). A core argument of contingency theory is that organizations would normally align their internal processes, structures, strategies, etc. with the prevailing external environment factors to obtain fit, and enhance performance outcomes (Donaldson, 2006; Johnson et al., 2001). The capability-based logic also maintains that firms' ability to respond to environmental changes largely depends on their distinctive capabilities, which can be enhanced through customer integration (Henderson & Clark, 1990; Hoopes & Madsen, 2008). From the arguments of contingency theory and capability-based view, we posit that competitor actions may drive supply chain responsiveness through customer integration.

When competitors engage in aggressive marketing activities, the focal firm is put under pressure to quickly retaliate using better marketing strategies, if they are to defend and secure their market position. Competitor price cuts would be met with drastic cost-cutting strategies through the downstream logistics function; fierce promotion would be met with growth strategies through extensive product availability or high market penetration; and product emulation would be met with increased R&D in order to frequently change the focal firm's offerings (Hutzschenreuter et al., 2021). Depending on the pace and number of competitors unleashing such aggressive initiatives, responding to these could overwhelm the focal firm. In order to effectively monitor competitor actions and efficiently respond to them, firms need to develop collaborations with its supply chain actors, particularly, in the downstream supply chain where competitor actions manifest. For instance, Peng et al. (2017) show that competitors' aggressive marketing behavior was responded to by insurers increasing their penetration strategies through bancassurance. Integrating processes with supply chain actors, therefore, allows for relevant information sharing, streamlining of processes for speed operations and harnessing of inter-organizational capabilities (synergistically) to amass the capacity to respond to competitor actions.

Similarly, when competitors frequently introduce new products or improve existing ones or their production technologies, the focal firm would need not only to increase their innovativeness and new product development but also the speed to market (Turner et al., 2010). Under such pressure, relying on internal resources and capabilities only may not avail much.

Consequently, the focal firm is more likely to draw on its relationship with supply chain partners to effectively and efficiently develop new products and processes to match the current competition demands. A parallel illustration is shown in the study of Moreira et al. (2020) where competitors' product launch compels global pharmaceutical firms to adopt technology licensing-in to quickly react to the market instead of undertaking their internal R&D to develop new products, which might be time-consuming. The point here is that under such desperate circumstances, firms tend to depend on inter-organizational capabilities than rely on their limited internal capability and resources.

Therefore, on the assumption that firms pursue rational decisions to maximize profit and survive, they cannot sit but respond to competitor actions with appropriate capabilities. For the focal firms to respond to competitor actions, and maintain their market position, there is the need to build their capabilities to improve business processes, product quality, and service delivery. This, we propose, can be achieved by the focal firm joining forces with supply chain partners (i.e., integration) to leverage complementary resources and competencies (Flynn et al., 2010; Munir et al., 2020). Consequently, it can be argued that competitor aggressiveness and competitor innovativeness can push focal firms toward integration with downstream partners in order to maximize their capabilities to respond to market needs. Also, as prior studies indicate, competitor actions motivate focal firms to collaborate strategically with their partners to manage inter-and intra-firm processes together (Flynn et al., 2010; Seo et al., 2014). Therefore, higher levels of competitor aggressiveness and competitor innovativeness are likely to increase the propensity of focal firms to engage in customer integration to benefit from external resources.

In turn, an integration between the focal firms and their downstream partners helps boost their operational capabilities from combined resources to enhance supply chain responsiveness. Largely, supply chain responsiveness thrives on effective information sharing, joint decision-making, and system-coupling mechanisms, which are key attributes of customer integration (Munir et al., 2020; Shou et al., 2018). More specifically, customer integration does not only allow channel partners to scan and pick market intelligence but also share information about sales forecasts, production plans, delivery status, and inventory to make them responsive to the market needs (Munir et al., 2020; Shou et al., 2018). As Flynn et al. (2016) indicate, accurate and timely information exchange

among supply chain actors can help reduce inventories, improve flexibility, and speed up production cycles. In addition, customer integration empowers partners to make joint and collaborative decisions on product and process designs, quality improvement, cost control and facilitates joint problem solving, particularly under situations of uncertainties to sustain and maximize operational excellence (Shou et al., 2018). Also, system-coupling with downstream partners, for example, vendor-managed inventory and continuous replenishment systems can help achieve the unified control of inter-firm processes, improve supply chain flexibility, and on-time delivery (Munir et al., 2020; Shou et al., 2018). As indicated in prior research, integrative partnership within supply chains is considered a critical source of capability that helps firms achieve mutually compatible goals and competitive advantages that they could not realize on their own (Srinivasan et al., 2020). Rather than local optimization or 'siloeing', customer integration recognizes that value can better be created by leveraging linked business processes through collaborative and coordinative efforts of channel actors (Annan et al., 2016; Frohlich & Westbrook, 2001).

In sum, we expect that competitor actions: aggressiveness, and innovativeness are likely to compel the firm to build customer integration, which in turn, drives supply chain responsiveness. The logic is that firms need to improve their competitive capabilities to deal with environmental uncertainties that emanate from competitor actions (Munir et al., 2020). Customer integration plays such a link mechanism. This expectation is supported by prior evidence that supply chain integration, particularly with downstream channel members, creates operational flexibility and allows firms to be responsive to market dynamics and customer changing needs (Kalyar et al., 2019; Munir et al., 2020; Shou et al., 2018). We accordingly hypothesize as follows:

H2a: Customer integration mediates the relationship between competitor aggressiveness and supply chain responsiveness.

H2b: Customer integration mediates the relationship between competitor innovativeness and supply chain responsiveness.

Methods and results

Source of data and field study

The study relied on a survey design and cross-sectional data. The researchers focused on firms

operating in the Ashanti Region of Ghana. Ashanti Region is one of the key administrative regions of Ghana. The focus on this context for the study was deemed relevant as the region constitutes an emerging market where new businesses are continuously springing up and thereby intensifying competition in its various business sectors (particularly, the manufacturing, the financial, and the hospitality sectors). The position of the region, being the center of the country, also allows it to possess key logistics and supply chain infrastructure that support several business and commercial activities. Broadly, two industries highly characterized by intense competition in the region were considered. These are the service industry (specifically, firms in the financial and hospitality sectors) and the manufacturing industry (i.e., food and beverages, pharmaceuticals and chemicals, agri-business, toiletries and cosmetics, and rubber, plastics and related).

To identify and locate the target firms we largely relied on the databases of the Yellow Pages Ghana and the Association of Ghana Industries to retrieve contacts and addresses of the firms. To ensure sample representation and minimize selection bias, we stratified the target population by industry and then assigned quotas to these subsamples based on the available information. As with the other parts of the country, the majority of the firms in the Ashanti Region of Ghana are in the service industry. For example, a recent survey of business establishments in the country reveals that approximately 82.7% and 16.8% of firms operate in the service and manufacturing industries, respectively (Ghana Statistical Service, 2016). Accordingly, and to ensure that the subsamples contain enough cases for statistical purposes, we purposively generated a sample of 366 firms, with 72.1% and 27.9% being service firms and manufacturing firms respectively. The proportion of these firm categories in our effective sample ($n = 117$) is similar: 72.6% are service firms while 27.4% are manufacturing firms. The firms sampled were those that met the criteria for the study: employed at least five full-time employees, have at least a form of integration with one or more channel members from the downstream of their supply chain, and with complete contact information.

In all, 170 questionnaires were hand-delivered to key informants who were willing to participate in the study. A total of 126 questionnaires were received in two phases. We retrieved 85 questionnaires in phase 1 (i.e., within two weeks of delivery of questionnaire) and 41 in phase 2 (i.e., after two weeks of delivery of questionnaire). Initial examinations of the

questionnaires received resulted in retaining 117 (i.e., 79 from phase 1, and 38 from phase 2) as effective for the study. Substantial missing values and incompleteness were the main reasons for rejecting 9 of the questionnaires received. T-test conducted on the composite scores of the variables of interest showed no statistically significant difference between phase 1 and phase 2 (Armstrong & Overton, 1977); given the following results: customer integration (*mean difference* = $-.35$, *t-value* = -1.456 , $p = .148$); competitor aggressiveness (*mean difference* = $-.24$, *t-value* = -1.317 , $p = .190$); competitor innovativeness (*mean difference* = $-.26$; *t-value* = -1.137 , $p = .258$); and supply chain responsiveness (*mean difference* = $-.117$; $t = -.700$; $p = .486$).

The average firm age was 19.72 years ($SD = 24.615$); and that of firm size (full-time employee) was 174 ($SD = 270.497$). In line with previous research (see e.g., Huo, 2012), for each firm contacted, one key informant who at least held a managerial position was identified to respond to the questionnaire. These individuals included supply chain/logistics manager (15.6%), CEO/president/vice president (30.3%), General Manager/Managing Director (19.3%), or other individuals perceived to have adequate knowledge about the company's internal and external processes (34.9%). The average number of years a key informant has held his or her current position was 3.85 years ($SD = 2.962$). Based on previous research approaches (see Boso et al., 2013) we assessed the competence of the informants along four dimensions using a 7-point scale ranging from 1 (*strongly disagree*) to 7 (*strongly agree*). The results obtained: knowledge adequacy on issues (*Mean* = 5.89; $SD = 1.120$), understanding of items (*Mean* = 5.95; $SD = 1.041$), confidence in responses (*Mean* = 6.09; $SD = .979$), and assurance of responses (*Mean* = 6.09; $SD = 1.059$) generally lent satisfactory assurance that the informants were competent enough to provide data for the study.

Questionnaire and measures

To improve face and content validity, we started developing the questionnaire for the study by first reviewing existing literature to help us identify and develop (where necessary) items and scales. In most cases, measures were adapted to suit the empirical context of the study. The questionnaire instrument was designed in the English language, the official language used in Ghana. To measure customer integration, we used a 7-point scale measuring from 1 (*not at all*) to 7 (*to the largest extent*) and adapted 7

measures from Flynn et al. (2010), Danese and Romano (2011), and Seo et al. (2014). The informants were asked to indicate the extent to which their firms have integrated with key downstream channel members along with the measures and the scale used. Using a scale of 1 (*strongly disagree*) to 7 (*strongly agree*), 3 measures each were used to measure competitor aggressiveness and competitor innovativeness. The items for the former construct were developed by the researchers while those for the latter were adapted from Boso et al. (2013) and Seo et al. (2014). The informants were asked to indicate the degree to which they agree or disagree on the extent to which each item prevails within their business environment. Lastly, relying on insights from Cadden et al. (2013), Chopra and Meindl (2007), and Miguel and Brito (2011), we identified 6 items to measure supply chain responsiveness using a 7-point scale, which ranged from 1 (*much worse*) to 7 (*much better*). In relation to a key competitor, we asked the informants to indicate how worse or better their firms can swiftly respond to the needs of their customers. In addition, we included other variables that could confound our model's prediction of the dependent variable. Literature has suggested that factors such as firm size and firm age/experience (Bourlakis et al., 2014) as well as firm industry (Um & Kim, 2019) may influence the capacity and the difficulty to effectively respond to customer demands in a dynamic market, respectively. Accordingly, we included firm industry ('0' = service, and '1' = manufacturing), firm age (years in operation), and firm size (number of full-time employees) as control variables following the examples of similar prior research (Jafari et al., 2022; Jajja et al., 2018; Yu et al., 2019, 2020) and prior studies from the empirical context (Boso et al., 2013; Danso et al., 2020).

Validity and reliability

Using MPLUS, an assessment of the measurement model was conducted using confirmatory factor analysis (CFA) with the maximum likelihood (ML) as the estimation method and covariance matrix as the input (Vickery & Vieira, 2011). We followed recommended measurement model specification and modifications steps (see Diamantopoulos & Siguaw, 2000; Hair et al., 2014), and a good model fit to data was obtained. The item loadings, *t-values*, composite reliability (CR), average variance extracted (AVE), and scale reliability results are shown in Table 1. The reliability of the scales was determined by primarily inspecting the Cronbach's alpha value and the CR.

Table 1. Measurement model results.

Constructs/measures/composite reliability (CR)/average variance extracted (AVE)	Standardised loadings	T values
Customer integration (Cronbach α = .875; CR = .877; AVE = .546)		
CI1: You and your key customers collaborate in planning and making decisions	0.715*	Fixed
CI2: You and your key customers have common interest and goals	0.886	8.768
CI3: You and your key customers provide supports and assistance to each other	0.784	8.190
CI4: Your business processes & systems are aligned with customers' needs	0.648	6.384
CI5: You and your key customers work toward building long-term relationships	^a	
CI6: You and your key customers share business information in real time	0.658	6.668
CI7: Material/service flows are aligned with your customers' purchase needs	0.715	7.258
Competitor aggressiveness (Cronbach α = .763; CR = .790; AVE = .564)		
CA1: Competitors use and rely on 'unhealthy' marketing campaigns (e.g., saying bad things about your firm or its products/services)	0.562	Fixed
CA2: Competitors continuously adapt to or 'copy' your processes & products	0.883	5.769
CA3: Competitors are always making efforts to win (your) customers	0.773	5.650
Competitor innovativeness (Cronbach α = .887; CR = .894; AVE = .740)		
CNOV1: Competitors are more concerned with products and process improvement	0.860	Fixed
CNOV2: Competitors continuously adapt to using better technology or methods	0.940	12.689
CNOV3: Competitors frequently introduce 'new' products and make modifications to existing ones	0.772	9.901
SC responsiveness (Cronbach α = .871; CR = .872; AVE = .532)		
SCR1: On-time delivery of products to customers	0.710	Fixed
SCR2: The time it takes to serve customers or fulfill customers' orders	0.744	7.417
SCR3: Flexibility & swiftness in meeting different customer needs	0.715	7.006
SCR4: The time it takes to adjust processes to meet changing market needs	0.748	7.337
SCR5: Swiftness in resolving process/service failure	0.764	7.501
SCR6: Swiftness in handling varied customer requirements	0.694	6.886

^aItem with poor loading; *All loadings significant at 1%.

The Cronbach's alpha values obtained (ranging from .763 to .887) were all above the minimum alpha value of .70 (Bagozzi & Yi, 2012) which implies that the retained items of the scales were internally consistent in measuring their respective constructs (Pallant, 2007; Vickery & Vieira, 2011). The internal consistency was further confirmed by computing the CR values for each construct; each of which turned out to be far greater than .70 (Bagozzi & Yi, 2012). The AVEs above .50 confirm an adequate average percentage of variance explained among the items representing each construct, and further confirm the measurement model's reliability. The measurement model also passed convergent validity, as the items loaded significantly (i.e., with *t*-values greater than 1.96) unto their respective constructs. The model's fit indices were within acceptable ranges: χ^2 (*df*) = 222.062 (146); non-normed Chi-square (χ^2/df) = 1.521 was less than 2.0; root mean squared error of approximation (RMSEA) = .067; non-normed fit index (NNFI) = .918; comparative fit index (CFI) = .930; and SRMR = .065 (Bagozzi & Yi, 2012; Hair et al., 2014). From Table 1, except for four items, all the items' loadings were above .70 which demonstrates that the measurement model's items correlated largely with their respective constructs, suggesting a satisfactory convergent validity (Hair et al., 2014; Vickery & Vieira, 2011).

To satisfy that the constructs in the study are distinct (i.e., discriminant validity), inspections were

made on the correlational results for each pair of constructs produced by the CFA model and compared with the square root of the AVEs (Fornell & Larcker, 1981). All the inter-construct correlation coefficients (see Table 2) were far below .729, the square root of the lowest AVE computed, which further suggests good discriminant validity (Hair et al., 2014). Again, the existence of moderate correlations (i.e., greater than .30) between constructs of the hypothesized paths lends support to the nomological validity of the proposed model (Hair et al., 2014).

Common method bias assessment

Since the data were collected from a single source and also were in the form of self-reports, there was a need to demonstrate that common method bias (CMB) is not a concern in this study (Podsakoff et al., 2003). At the instrument design stage, various procedural remedies as suggested by Podsakoff et al. (2003) were followed. First, different scale anchors were employed to measure the constructs. Second, we separated the predictor and criterion variable in the survey instrument by interspersing measures of other constructs (not used in this study) in between them to reduce consistency motif and illusionary correlation (Podsakoff et al., 2003). Statistically, in MPLUS, Herman's one-factor model was constructed by allowing all items to load onto a single latent factor. The results obtained: χ^2/df = 716.830/152 = 4.716, RMSEA = .178, NNFI =

Table 2. Inter-construct correlations (CFA) and descriptive statistics results.

Variables	1	2	3	4	5	6	7
1. Competitive aggressiveness	0.751						
2. Customer innovativeness	0.414**	0.860					
3. Customer integration	0.310**	0.413**	0.739				
4. Supply chain responsiveness	0.184*	0.175	0.357**	0.729			
5. Firm age	0.006	0.098	0.003	0.218*			
6. Firm size	0.121	0.216*	0.034	0.147	0.747**		
7. Industry (manufacturing = 1)	-0.187*	-0.286**	0.030	-0.128	0.106	0.032	
Mean	4.658	4.903	4.759	4.913	2.454	4.216	0.274
Standard deviation	0.901	1.161	1.095	0.836	1.033	1.464	0.448
Variance inflation factor [†]	1.272	1.545	1.295	n/a	2.325	2.405	1.160

Notes: * $p < .05$; ** $p < .01$; [†]Dependent variable is supply chain responsiveness; n/a = not applicable; Boldface coefficients in the principal diagonal are square root values of the AVE for the respective constructs.

.416, $CFI = .482$, $SRMR = .147$, showed poor model fit; suggesting that our proposed multi-factor measurement model (i.e., the CFA discussed earlier) was far superior. Hence, the threat of common method bias is not a pervasive issue in the current study.

Structural model estimation and hypothesis evaluation

We first examined the correlational results to see if it was even necessary to proceed with estimating the proposed model. The results in Table 2 show relevant positive associations between both dimensions of competitor actions and customer integration as well as supply chain responsiveness. Also, a moderate positive association was found between customer integration and supply chain responsiveness. Proceeding with these results, we used the MPLUS statistical package to structurally estimate the proposed direct paths and the indirect (mediation) paths in the study's model, controlling for firm industry (manufacturing = 1; service = 0), firm age (natural log), and firm size (natural log). The output summarized in Table 3 shows the unstandardized coefficient (β) for each path, along with the respective standard errors (S.E) and p -values (significance). For the endogenous variables (i.e., customer integration and supply chain responsiveness), the coefficient of determination (r-square) is reported. The overall structural model shows acceptable fit to the dataset: $\chi^2/df = 269.995/180 = 1.500$, $RMSEA = 0.065$, $NNFI = 0.903$, $CFI = 0.916$, $SRMR = 0.075$.

The results in Table 3 show that competitor aggressiveness and competitor innovativeness significantly account for 21.2% of the total variance in customer integration. The overall model explains 24.2% of the total variance in supply chain responsiveness. Among the control variables, only firm age has a significant positive relationship with supply chain responsiveness ($\beta = .245$; $p < .05$). Both industry type ($\beta = -.339$; $p > .05$), and firm size ($\beta = -.039$; $p > .05$) have no statistical significance. Contrary to our expectation,

the direct paths from competitor aggressiveness and competitor innovativeness to supply chain responsiveness were each not significant: ($\beta = .060$; $p > .05$) and ($\beta = -.059$; $p < .05$), respectively. Thus, the results do not support hypotheses H1a and H1b. On the other hand, customer integration is found to have a significant positive relationship with SCR ($\beta = .325$; $p < .01$). Additional outputs on the indirect effects of the model from the MPLUS package are reported under the section "Indirect Effects" in Table 3. The results show a significant positive indirect relationship between competitor aggressiveness and supply chain responsiveness through customer integration ($\beta = .130$; 95% BCI: .023–.324), which supports hypothesis H2a. Similarly, the indirect association between competitor innovativeness and supply chain responsiveness through customer integration is significantly positive ($\beta = .093$; 95%BCI: .039–.187), which also provides support for hypothesis H2b.

Discussions and implications

The study drew on the contingency theory and capability-based view to examine how competitor actions affect supply chain capability development and performance outcome. Specifically, the question of whether or not competitor aggressiveness and competitor innovativeness directly affect supply chain responsiveness, and indirectly through customer integration was addressed empirically. Our theory and findings offer theoretical and managerial implications.

Contrary to this study's expectation, results did not support a direct relationship between competitor actions and supply chain responsiveness. A plausible explanation may be that competitor actions can stun or confuse focal firms and make them disorganized in their operations, at least in the short term, as they may not be adequately resourced or well-positioned to respond to competitive actions. In addition, firms' responses to competitive actions may take time to yield the desired results. As such, although focal firms

Table 3. Structural model analysis.

Direct effects Independent variables:	Customer integration			Supply chain responsiveness (SCR)		
	β	S.E.	p-value	β	S.E.	p-value
Industry-type (manufacturing = 1)				−0.339	0.223	.129
Firm size				−0.039	0.065	.550
Firm age				0.245	0.102	.016
Competitor aggressiveness (CA)	0.400	0.238	.094	0.060	0.223	.789
Competitor innovativeness (CINNV)	0.285	0.115	.013	−0.059	0.082	.471
Customer integration (CI)				0.325	0.111	.003
<i>Model fit indices:</i>						
χ^2/DF				269.995/180 = 1.500		
RMSEA				0.065		
NNFI				0.903		
CFI				0.916		
SRMR				0.075		
R ²	0.212				0.242	
p-value of R ²	0.008				0.009	
INDIRECT EFFECTS						
Indirect path		Indirect effect		95% Bootstrap confidence interval*		
CA → CI → SCR		0.130		0.023, 0.324		
CINNV → CI → SCR		0.093		0.039, 0.187		

Notes: Unstandardized estimated are reported; *5000 bootstrap samples.

would naturally be expected to react to competitor actions, such responses may not be effective enough (at least in the immediate term) as they may be born out of panic, and without a well-thought-through strategy (Setiawan et al., 2015). To counteract competitor actions and remain responsive to market needs, the focal firm needs to carry out more actions quickly, which may require external resources and interventions, such as forming integrative partnerships to improve resources and capabilities. The finding departs from the study of Ibrahim and Harrison (2020) who found a strong relationship between competitor action and overall business performance. It, however, finds consistency with the prior assertion that responding to market turbulence without commensurate capabilities and resources may prove futile (Setiawan et al., 2015). These findings, although unexpected, provide insight by showing that firms need to develop sufficient capabilities and resources to improve their responsiveness in the midst of competitor aggressiveness and competitor innovativeness (Hutzschenreuter et al., 2021).

A further theoretical implication of this study is that the competitor actions-supply chain responsiveness relationship can be viewed as significantly influenced by customer integration. Supporting our core argument, the results indicate that customer integration mediates the relationship between competitor actions and supply chain responsiveness. Grounded in the environment-structure/strategy-performance view of the contingency theory and the capability-based view, we developed and tested a nomological framework that argues that competitor aggressiveness and competitor innovativeness influence customer

integration, which in turn, drives supply chain responsiveness and obtained sufficient statistical support. This outcome provides a fine-grain understanding of how competitor actions influence supply chain responsiveness and helps highlight the bright side of competitor actions. Prior studies have paid less attention to competitor innovativeness and its influence on the focal firm's strategic choices. Largely, previous studies have focused on the innovativeness and innovation-related issues of the focal firm rather than that of competitors (Ibrahim & Harrison, 2020). It is worth acknowledging that firms operate in open systems that exert considerable influence on their operations and performance outcomes. As such, giving inadequate attention to what competitors are doing could be fatal (Hutzschenreuter et al., 2021; Setiawan et al., 2015). This study contributes to the existing literature by demonstrating how firms' perceptions of competitor innovativeness in their operating environment stimulate their integrative initiatives and consequently, responsiveness. A case is made here that, observing, and giving attention to competitors' innovative behaviors pay off for supply chains.

Furthermore, competitor aggression in many industries may be manifested in the form of price wars, unfriendly advertisement campaigns/promotions, copying the focal firm's strategies, business model/processes, products, etc. On the other hand, competitor innovativeness may be reflected in cutting down on cost, developing new products, improving business processes, extending product life cycle, speeding time to market, enhancing product availability, etc. (Vickery et al., 2003), in order to maintain or offer

better value for the customer. At the individual firm level, such pressures from the environment can be overwhelming and nearly unsurmountable in solo. Leveraging the synergetic advantage of channel members (through integrative efforts by way of sharing innovative ideas, customer information, and other resources) becomes a crucial element in withstanding such environmental turbulence (Boso et al., 2013; Luo et al., 2008; Moliterno & Mahony, 2011).

As revealed by the study's results, competitor aggressiveness and competitor innovativeness have significant positive effects on the customer integration of the supply chain in the study's context, which in turn enhances supply chain responsiveness. These findings suggest that in a competitive market, characterized by highly competitive pricing strategies, aggressive sales promotions, price wars, and innovation, customer integration may no longer be an option for market players. More specifically, since competitor action-induced uncertainties largely manifest in the downstream supply chain, customer integration becomes particularly imperative for businesses to respond appropriately to such environmental turbulence, even for survival (Boso et al., 2013; Flynn et al., 2016).

Also, results show that customer integration, in turn, drives supply chain responsiveness. Supply chain integration within the downstream is often effective in making supply chains responsive to marketing needs following market turbulence upsurge caused by competitors (Flynn et al., 2016). By implication, this result suggests that, given high levels of competitor aggressiveness and competitor innovativeness, firms tend to develop unique capabilities through alignment of their downstream resources and engagement in other integrative initiatives (e.g., information sharing, collaboration, coordination) with downstream channel members to wield competitive advantage amidst market turbulence. It is through such integrative efforts that a firm could marshal compelling strength to meet market turbulence following competitor actions and swiftly address market needs to possibly stay competitive. Thus, an effective response to market turbulence, such as competitor actions, necessitates a capability that provides the possibility of an enhanced supply chain. Customer integration fulfills this role.

Finally, as the results indicate, firm age has a positive significant influence on supply chain responsiveness. The results suggest that firms' experience in the current market matters in achieving responsiveness in the face of competitor actions. In particular, the years of experience implies knowledge of the competitive

dynamics and their implications better. On the other hand, while the firm size is believed to enhance a firm's awareness of opportunities and its capability to influence the competitive environment (Hughes-Morgan et al., 2018), the results suggest the contrary. This may be because firm size may reduce a firm's capability to act flexibly since size can lead to the institutionalization of firm behaviors and routines, and thus, makes responsiveness difficult (Hughes-Morgan et al., 2018). Similarly, and surprisingly, the results did not offer support for the influence of industry type on supply chain responsiveness. Given the differences in the supply chain structure of the two industries (i.e., manufacturing and service), we expected supply chain responsiveness to differ by industry. However, the lack of significant difference in supply chain responsiveness between the two industries could be accounted for by micro-level socioeconomic factors influencing the industries by the same or similar magnitude.

Managerial and policy implications

We emphasize that a lack of integration in the value creation process could lead to suboptimization of results (Annan et al., 2016), particularly in a competitive environment where swift responses to environmental turbulence and customer demands are imperative. Customer integration has the potential to provide management with operational flexibility to respond rapidly to market dynamics such as competitor actions and changes in customer needs, demands, and preferences (Annan et al., 2016; Awad & Nassar, 2010; Stonebraker & Liao, 2004; Vickery et al., 2003). In competitive environments, market intelligence is so crucial as it helps firms to understand the activities of competitors and the changing needs of customers. However, without integrating with downstream actors, obtaining such intelligence could be constrained. Readiness and willingness of downstream channel members who usually possess such intelligence to share information depend on the extent to which the focal firm integrates with them (Flynn et al., 2016).

Based on these findings, customer integration is one of many capabilities linking competitor actions and supply chain responsiveness. We take the view that markets characterized by high levels of competitor aggressiveness and competitor innovativeness can result in a coalition built within supply chains to enhance performance. By implication, it can be said that in markets where no or low levels of competitor aggressiveness and competitor innovativeness are

experienced, firms may become dormant, lack the drive to build capabilities within their supply chains, and ultimately experience lower supply chain responsiveness. Therefore, efforts by governments and other pressure groups to liberalize industries or loosen barriers to competition should be encouraged. Also, industry players or managers should welcome market turbulence caused by competitive forces, take advantage of the opportunities that it offers and adopt strategies that minimize any threat thereof to enhance business performance.

Conclusion, limitations, and avenues for further studies

The study sought to contribute to the scant literature on the conundrum between competitor action (a factor of environmental uncertainty) and supply chain responsiveness. Prior studies have acknowledged that competitor actions may have a threat-opportunity paradox effect on rival firms' performance. However, theoretical and empirical evidence for this assertion is lacking. The study contributes to resolving this paradox by analyzing data from Ghana, a Sub-Saharan African economy. The findings suggest that each of the two competitor actions examined (i.e., competitor aggressiveness and competitor innovativeness) has no significant direct relationship with supply chain responsiveness, but only through customer integration—an inter-organizational capability variable. The study, therefore, contributes to resolving the threat-opportunity paradox effect of competitor action on firm performance, by showing that customer integration is one of several mechanisms through which the bright side (opportunity) of competitor actions can be realized. From the supply chain perspective, we conclude that competitor aggressiveness and competitor innovativeness may not be necessarily bad.

Regardless of the important contributions from the study, it is not without limitations. The current study limited its scope to only two environmental turbulent factors (i.e., competitor aggressiveness and competitor innovativeness). We acknowledge that there may be several other factors such as market saturation by new entrants, governmental and legal provisions causing businesses to operate in a more stringent environment (policies on tax, environmental emissions, controlled economy, import/export policies, etc.), technological disruptions, etc. that could be a toll on businesses to marshal forces to respond to such turbulent factors accordingly. We, therefore, suggest that future studies incorporate some of these factors in modeling

environmental turbulence and assess how they may affect supply chain integration and responsiveness.

Secondly, the study only focused its attention on how environmental turbulence affects supply chain responsiveness through customer integration. However, competitor aggressiveness, for instance, may influence the entire supply chain other than just the downstream. Typically, competitors engaging in price wars and fierce sales promotional strategies may require integrative efforts from various actors within the successive stages of the supply chain; that is, in the form of procurement of quality and cost-effective resources to just-in-time delivery of stocks, etc. Similarly, depending on the level of competitor innovativeness, an integrated effort from raw materials suppliers through the focal firm to distributors may be required to respond to market disruptions. These lend credence to the fact that competition has moved from among firms to among supply chains (Christopher, 2011). As such, it takes an integrative effort across the entire supply chain to build the competencies necessary to meet the turbulent environment. The breadth of integration across the supply chain is found to correlate significantly with performance (Childerhouse & Towill, 2011; Flynn et al., 2010; Frohlich & Westbrook, 2001). Given this, it is recommended that future studies focus on examining how the broader scope of integration serves as a conduit for transmitting the effects of competitor actions on performance.

Again, we suggest further studies to consider the interaction or cumulative effects of environmental factors on supply chain integration and performance outcome. For instance, the correlation result in this study suggests that aggressive competitors are also innovative. Essentially, significant aggression from competitors also implies that they are innovative in their competitive strategies. Therefore, exploring how these variables cumulatively or jointly affect supply chain practices and performance outcomes would be insightful. Moreover, the results show that the effect of competitor innovativeness on customer integration was comparatively stronger than that of competitor aggressiveness. We infer that an aggressive competitive action may not be a sufficient driver for integration, especially in environments or industries where the rate of innovation is insignificant. By a logical implication, aggressive competitive actions may likely result in just a retaliating aggressive response without necessarily driving integration. Relying on a larger sample and comparing the study's model in different industries which differ in innovation pace would be insightful in providing clarity on this argument.

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